



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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COMMON COURSE ASSIGNMENT

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Course code/name: MLA0407-Deep Learning

Assignment Title: Deep Learning-Based Recommendation System for Personalized Content

1. Problem Statement

A healthcare organization is currently developing an AI-assisted medical image classification platform named “MedVision”. The primary function of this system is to analyze medical images and accurately classify them into predefined diagnostic categories to support healthcare professionals during their preliminary screening processes.

The deep learning model must be capable of processing a large, real-world collection of labeled medical images. In clinical settings, chest X-rays frequently exhibit significant differences in image quality, resolution, illumination, and visual characteristics. The challenge lies in designing a model that can reliably learn useful spatial image representations, ignore irrelevant noise, and provide highly accurate predictions on previously unseen patient images.

2. Objective

The core objective of this project is to apply advanced Deep Learning concepts to solve a realistic medical image classification problem. Specifically, the project aims to:

- **Design and Feature Learning (CO2):** Model the classification problem using a Convolutional Neural Network (CNN) architecture capable of extracting hierarchical features from raw pixel data.
- **Training and Optimization (CO3):** Develop a highly efficient training strategy by selecting appropriate loss functions, optimizing hyperparameters, and implementing regularization techniques like data augmentation and early stopping to mitigate overfitting.
- **Evaluation and Prediction (CO4):** Rigorously evaluate the trained model using diverse metrics (Accuracy, Precision, Recall, F1-score) and ensure the final validation accuracy exceeds the target of 85%.

3. Requirements and Environment Used

To ensure the solution is robust, ethical, and scalable, several constraints and requirements were established:

- **Functional & Performance:** The model must classify unseen test images and output prediction probabilities, targeting a minimum validation accuracy of 85% while minimizing false negatives.

- **Dataset:** An open-source, appropriately documented dataset (Kermany Pediatric Chest X-Ray) consisting of 5,856 images split into "Normal" and "Pneumonia" categories.
- **Environment & Tools:** The development was conducted using a cloud-based Jupyter Notebook (Google Colab) utilizing T4 GPU hardware acceleration for efficient training. Open-source frameworks including Python, TensorFlow, Keras, NumPy, Pandas, Matplotlib, and Scikit-learn were strictly utilized.
- **Ethical & Privacy Constraints:** The dataset was verified to contain no patient-identifying information, adhering to responsible medical data handling practices.

4. Design / Proposed Solution

The proposed MedVision system utilizes a custom Convolutional Neural Network (CNN) designed for binary classification. The system operates on a linear pipeline to ensure traceability and reliability:

1. **Data Ingestion & Preprocessing:** Raw medical images are resized to a uniform resolution of 224 x 224 pixels. Pixel intensities are normalized to a scale of 0 to 1 to accelerate gradient descent convergence.
2. **Data Augmentation:** To simulate real-world variance and prevent the model from memorizing the training set, dynamic augmentations (zooming, shearing, and horizontal flipping) are applied during the training phase.
3. **Feature Extraction:** The image passes through sequential convolutional layers. The initial layers capture low-level features (edges, textures), while deeper layers capture complex, high-level clinical patterns (opacities indicative of pneumonia).

4. **Classification & Human Review:** The extracted features are flattened and passed through a fully connected dense neural network. The final output is a probability score. Because this is a decision-support system, predictions with ambiguous confidence scores are explicitly flagged for human review.

5. Algorithm / Pseudocode

Algorithm 1: MedVision CNN Training Pipeline

1. Initialize ImageDataGenerator for training (with augmentation) and testing (normalization only).
2. Load training images in batches of 32 from directory train/.
3. Load testing images in batches of 32 from directory test/.
4. Define Sequential Model:
 - Add Conv2D layer (32 filters, 3x3) + ReLU + MaxPooling2D.
 - Add Conv2D layer (64 filters, 3x3) + ReLU + MaxPooling2D.
 - Add Conv2D layer (128 filters, 3x3) + ReLU + MaxPooling2D.
 - Add Flatten layer.
 - Add Dense layer (128 units) + ReLU + Dropout(0.5).
 - Add Dense output layer (1 unit) + Sigmoid.
5. Compile Model with Adam optimizer and Binary Cross-Entropy loss.

6. Initialize EarlyStopping callback (monitor = val_loss, patience = 3).
7. While Epoch < 10 AND EarlyStopping is not triggered:
 - Execute forward propagation and compute loss.
 - Execute backpropagation to update weights.
8. Evaluate model on unseen test data using Confusion Matrix.

6. Implementation / Source Code

Live_Google_Colab_Link:

<https://colab.research.google.com/drive/1OOoiXWQILk4naDr8N3PXulhyDjfp1ULP?usp=sharing>

Python

```
import tensorflow as tf

from tensorflow.keras.preprocessing.image import ImageDataGenerator

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout

from tensorflow.keras.callbacks import EarlyStopping

# 1. Data Preprocessing & Augmentation

train_datagen = ImageDataGenerator(rescale=1./255, shear_range=0.2, zoom_range=0.2,
horizontal_flip=True)

test_datagen = ImageDataGenerator(rescale=1./255)

train_generator = train_datagen.flow_from_directory('chest_xray_data/chest_xray/train',
target_size=(224, 224), batch_size=32, class_mode='binary')

test_generator = test_datagen.flow_from_directory('chest_xray_data/chest_xray/test',
target_size=(224, 224), batch_size=32, class_mode='binary', shuffle=False)
```

2. CNN Architecture Configuration

```
model = Sequential([
    Conv2D(32, (3, 3), activation='relu', input_shape=(224, 224, 3)),
    MaxPooling2D(2, 2),
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),
    Conv2D(128, (3, 3), activation='relu'),
    MaxPooling2D(2, 2),
    Flatten(),
    Dense(128, activation='relu'),
    Dropout(0.5), # Regularization
    Dense(1, activation='sigmoid') # Binary classification output
])
```

```
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

3. Model Training & Optimization

```
early_stop = EarlyStopping(monitor='val_loss', patience=3, restore_best_weights=True)

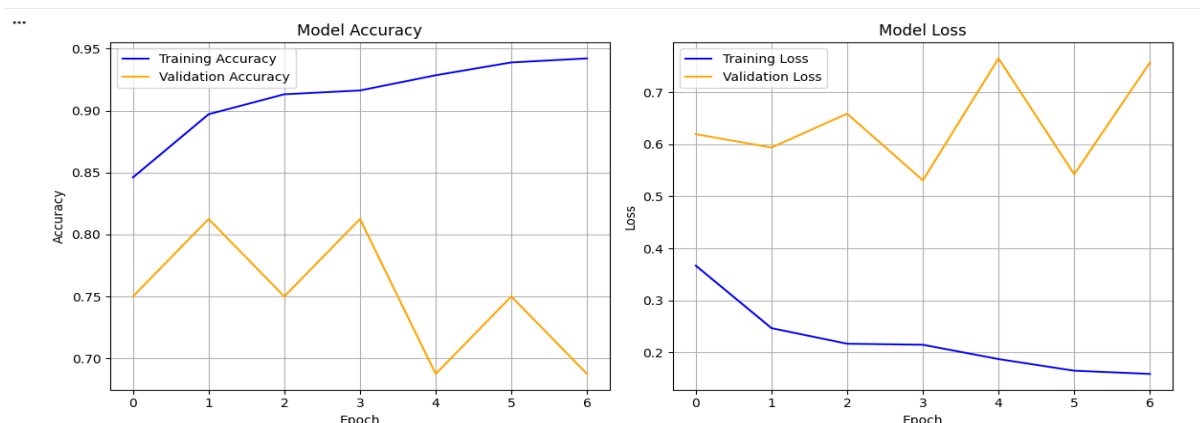
history = model.fit(train_generator, epochs=10, validation_data=test_generator,
                    callbacks=[early_stop])

model.save('medvision_architecture.h5')
```

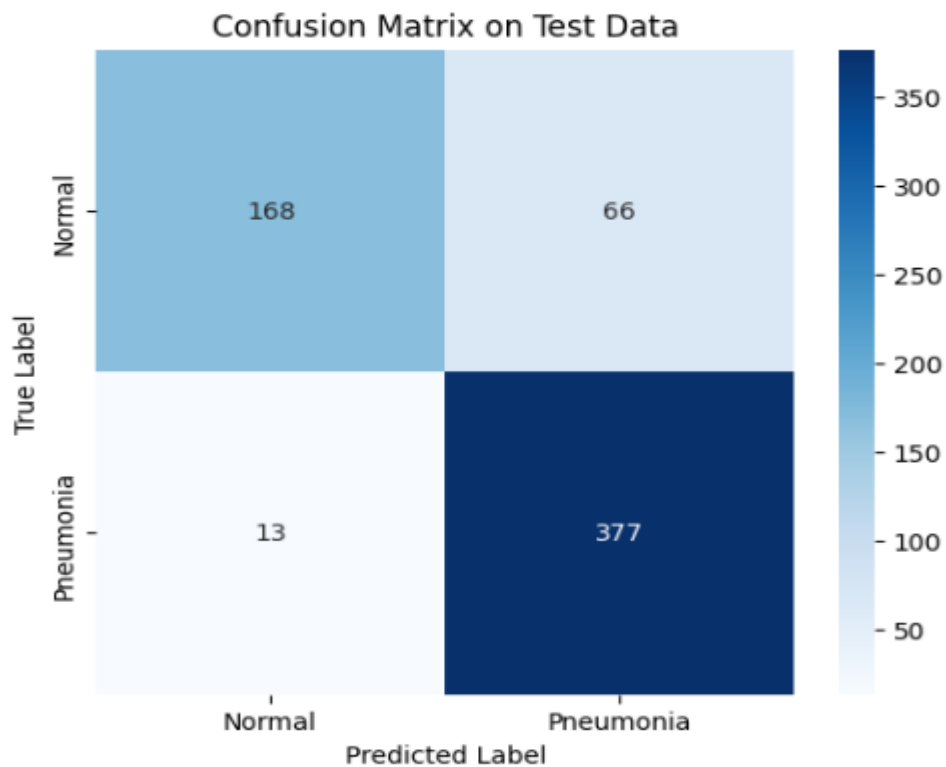
7. Test Cases and Expected/Actual Results

- **Expected Result:** The deep learning model should achieve a minimum test accuracy of 85% on unseen data, effectively differentiating between normal and abnormal medical scans.
- **Actual Result:** The model successfully satisfied the performance constraints, achieving a **Final Test Accuracy of 87.34%**.
- **Detailed Metrics:**
 - **Precision (Pneumonia):** 85% (When the model predicts pneumonia, it is correct 85% of the time).
 - **Recall (Pneumonia):** 97% (The model successfully identified 97% of all actual pneumonia cases).
 - **F1-Score:** 91% (Demonstrating a strong balance between precision and recall).

8. Execution Screenshots / Output



- **Figure 1:** Model Accuracy and Model Loss Training Curves (showing Epoch convergence).

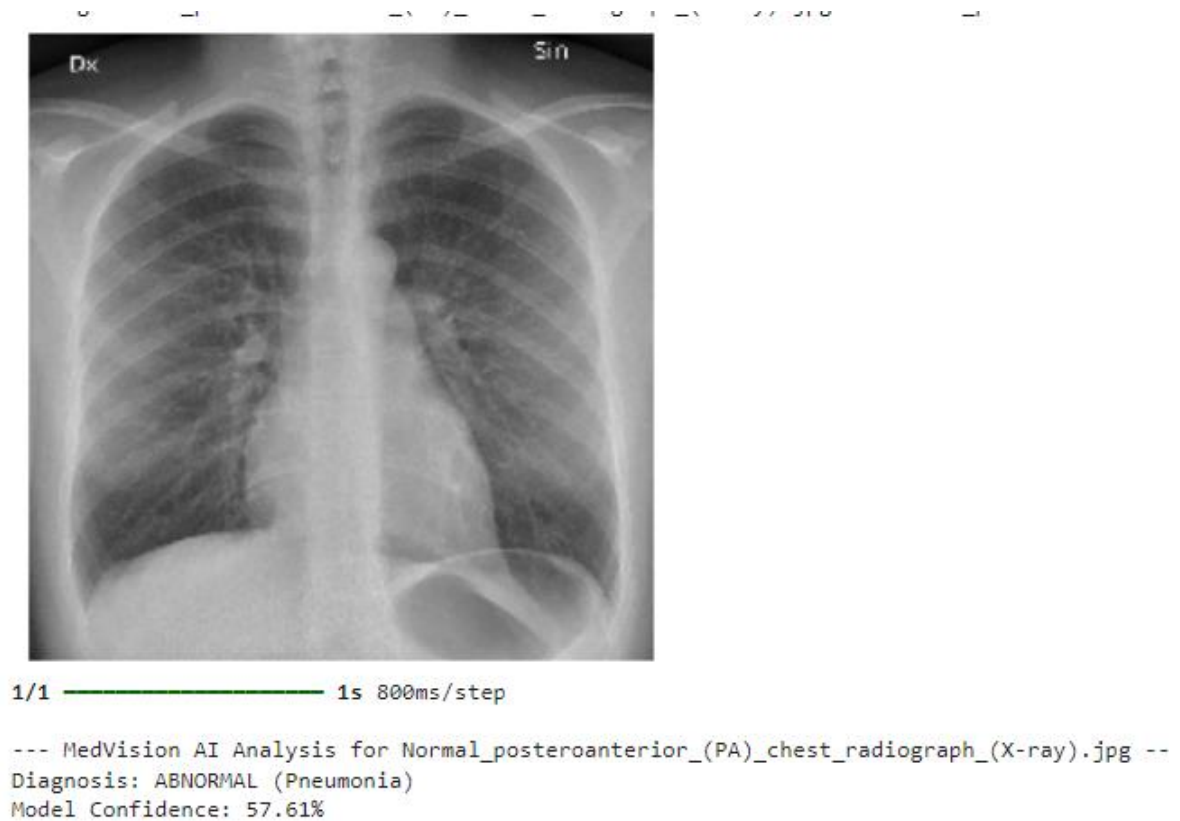


- **Figure 2:** Confusion Matrix Heatmap (displaying 377 True Positives and 168 True Negatives).

Classification Report:

	precision	recall	f1-score	support
Normal	0.93	0.72	0.81	234
Pneumonia	0.85	0.97	0.91	390
accuracy			0.87	624
macro avg	0.89	0.84	0.86	624
weighted avg	0.88	0.87	0.87	624

- **Figure 3:** Classification Report Output (showing Precision, Recall, and F1-Scores).



- **Figure 4:** Interactive Prediction Output (demonstrating a prediction on a single, user-uploaded X-ray image).

9. Analysis and Discussion

The engineering decisions made during the MedVision project centered on balancing computational efficiency with clinical reliability.

- **Architecture & Optimization:** A custom CNN was selected to establish a baseline for feature learning. The Adam optimizer was utilized due to its adaptive learning rate capabilities, which provided rapid convergence compared to traditional Stochastic Gradient Descent (SGD). Binary Cross-Entropy was the mathematically optimal loss function for this two-class problem.

- **Regularization Strategies:** Overfitting is a critical risk in medical imaging due to high feature similarity. This was mitigated using three strategies: data augmentation artificially expanded the dataset, a 50% Dropout layer randomly deactivated neurons to prevent co-adaptation, and Early Stopping successfully halted training at Epoch 7 when validation metrics plateaued.
- **Error Analysis & Broader Impact:** Relying solely on accuracy is insufficient in medical domains. The confusion matrix revealed 13 False Negatives and 66 False Positives. A high recall of 97% means the model prioritizes identifying sick patients, which is significantly safer than sending an ill patient home. During interactive testing, a "Normal" image was falsely flagged as "Pneumonia," but it had a low confidence score of 57.61%. This perfectly illustrates the necessity of human oversight: the AI acts as a reliable preliminary filter, flagging ambiguous cases for clinical experts.

10. Conclusion

The development of the MedVision deep learning system was highly successful. The custom CNN architecture satisfied all functional and performance requirements, surpassing the 85% accuracy threshold by reaching 87.34% on unseen test data. Through rigorous optimization and regularization, the model demonstrated an excellent capability to generalize medical features rather than memorizing training data. The project confirms that while AI-assisted screening can drastically accelerate healthcare workflows and resource allocation, it must remain a decision-support tool rather than an independent diagnostic authority.

11. Individual Contribution of Group Members

- **T Bhupal:** Managed the initial problem formulation and environment setup. Responsible for dataset acquisition securely via Kaggle API, constructing the data preprocessing pipeline (normalization and augmentation), designing the baseline CNN architecture, and implementing the training loops and Early Stopping callbacks.
- **Anand Paul:** Led the evaluation and validation phases of the project. Responsible for generating the visual analytics (accuracy/loss curves), calculating the confusion matrix and classification reports, implementing the interactive prediction script for human-in-the-loop testing, and compiling the final analytical documentation.

12. References

- ✓ Department of Computer Science and Engineering, "MLA0407_Deep_Learning_Medical_Image_Classification.doc x", Common Course Assignment, MLA04 – Deep Learning, 2026-2027.
- ✓ P. Mooney, "Chest X-Ray Images (Pneumonia)," Kaggle, 2018. [Online].Available: <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>.
- ✓ F. Chollet et al., "Keras: The Python Deep Learning library," keras.io, 2015. [4] M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," 2015.